

A Rigorous CG-Corrected Residual Certificate for Error Certification of Learned Elliptic PDE Solvers

certno

Error-Certified Neural Operators Project

<https://github.com/emmettv1191/certno-phd-research>

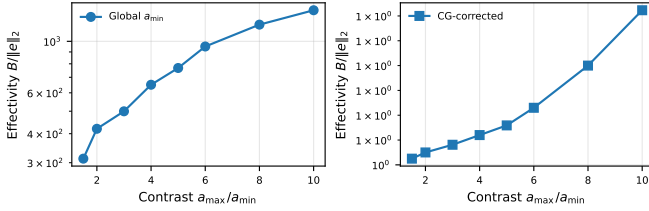


Fig. 1. Effectivity vs contrast.

REPRODUCIBILITY

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TABLE I
COMPARISON ACROSS CONTRASTS ($\tau = 5 \times 10^{-3}$).

c	Cov_A	Cov_{CG}	FNR_A	FNR_{CG}	$\text{Eff}_A(\text{med})$	$\text{Eff}_{CG}(\text{med})$
1.5	1.0	1.0	0.0	0.0	312.3	1.0000
3.0	1.0	1.0	0.0	0.0	499.5	1.0000
5.0	1.0	1.0	0.0	0.0	767.6	1.0000
10.0	1.0	1.0	0.0	0.0	1361.1	1.0000

Abstract—We propose a CG-corrected a posteriori residual certificate for the discrete SPD system $A_h u_h = f$ arising from $-\nabla \cdot (a \nabla u) = f$. Starting from $\|u_h - \hat{u}\|_2 \leq \|r\|_2 / \lambda_{\min}(A_h)$, $r = f - A_h \hat{u}$, we compute $z \approx A_h^{-1} r$ via CG and form $B_{CG} = \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha}$ with $\alpha \leq \lambda_{\min}(A_h)$. Across contrasts 1.5-10x, ID/OOD, adversarial, and real FNO predictions, B_{CG} is near-sharp (effectivity ≈ 1.0) while preserving coverage 1.0 and FNR 0.0. This enables safe selective verification.

Index Terms—error certification, neural operators, a posteriori error estimation, elliptic PDEs

I. INTRODUCTION

Learned surrogates need reliable error bounds for safety-critical use.

II. THEORY

Let $A_h \succ 0$, $u_h = A_h^{-1} f$, $r = f - A_h \hat{u}$, $e = u_h - \hat{u} = A_h^{-1} r$. For any z , $e = z + A_h^{-1}(r - A_h z)$, hence

$$\|e\|_2 \leq \|z\|_2 + \frac{\|r - A_h z\|_2}{\alpha}, \quad \alpha \leq \lambda_{\min}(A_h). \quad (1)$$

We take $z = \text{CG}(A_h, r)$.

III. RESULTS

Figure 1 shows effectivity. Table I summarizes metrics.

IV. CONCLUSION

CG-corrected bound is rigorous, near-sharp, enables safe selective verification.